

DATA CLEANING AND TRANSFORMATION OBSERVATION

Overview:

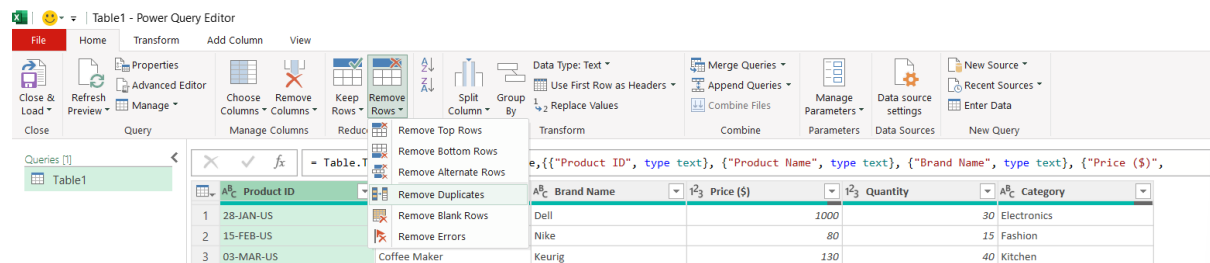
This project showcases data cleaning and transformation in Excel Power Query, including duplicate removal, data standardization, missing value handling, and dataset preparation for analysis.

Observation workflow:

Loaded the data to Power Query through **Data-> Get data-> from Table /Ranges**

1. Removing Duplicates:

Removed the duplicates by selecting the column Product ID which has most unique values and options in query, **Remove rows-> Remove duplicates**



2. Correcting the inconsistent Data:

- Identified the incorrect data in the column Category and Product name by filtering the columns
 - ✓ Inconsistent data- Electroni, laptop, smartphone, headphone
- Corrected the inconsistent data into valid one by selecting the related columns **Transform-> Replace Values**

Table1 - Power Query Editor

File Home Transform Add Column View

Group By Use First Row as Headers Count Rows

Data Type: Text * Replace Values * Unpivot Columns * Merge Columns * Statistics Standard Scientific Rounding Information * Date * Time * Duration * Expand * Aggregate * Extract Values * Create Data Type

Reverse Rows Detect Data Type Rename Pivot Columns Convert to List Split Column * Format * Parse * Text Column Number Column Date & Time Column Structured Column

Queries [1] Table1

Table.ReplaceValue(#"Removed Duplicates", "Electroni", "Electronics", Replacer.ReplaceValue, {"Category"})

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
1	28-JAN-US	Laptop	Dell	1000	30
2	15-FEB-US	Sneaker			Electronics
3	03-MAR-US	Coffee			null
4	11-APR-US	smartp			null
5	22-MAY-US	Backpak			null
6	07-JUN-UK	Headph			null
7	19-JUL-UK	T-shirt			null
8	23-AUG-UK	Blender			null
9	05-SEP-UK	Tablet			null
10	14-OCT-UK	Hiking B			null
11	17-JUN-IN	Laptop			null
12	25-NOV-AU	Sneaker			null
13	08-DEC-DE	Coffee			null
14	18-FEB-CA	Smartw			null
15	16-APR-ES	headph			null
16	21-AUG-CA	Laptop			null
17	20-AUG-CN	Smartw			null
18	27-JAN-IT	Laptop			null
19	01-MAR-UK	Sunglasses	Oakley	250	15
20	14-AUG-US	Camping Tent	Coleman	null	10
21	14-MAY-RU	Camera	Nikon	700	50
22	09-JAN-CA	Microwave	Panasonic	80	20
23	19-JUL-BR	Fitness Tracker	Xiaomi	250	30
24	29-SEP-CA	Smartphone	Google	800	45
25	03-JUN-CA	Sunglasses	Ray-Ban	null	25
26	11-JUL-CA	Blender	Vitamix	400	40
27	07-MAR-CA	Dress	Zara	60	30
28	13-APR-CA	Toaster	Hamilton	40	10

Replace Values

Replace one value with another in the selected columns.

Value To Find: Electroni

Replace With: Electronics

Advanced options

☒ Match entire cell contents

☐ Replace using special characters

Insert special character

OK Cancel

Query Settings

PROPERTIES

Name: Table1

APPLIED STEPS

Source

Changed Type

Removed Duplicates

Replaced Value

Table1

Table.ReplaceValue(#"Corrected the Product name", "smartphone", "Smartphone", Replacer.ReplaceValue, {"Product Name"})

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
1	28-JAN-US	Laptop	Dell	1000	30
2	15-FEB-US	Sneakers	Nike	80	15
3	03-MAR-US	Coffee Maker	Kuorini	120	40
4	11-APR-US	smartp			null
5	22-MAY-US	Backpak			null
6	07-JUN-UK	Headph			null
7	19-JUL-UK	T-shirt			null
8	23-AUG-UK	Blender			null
9	05-SEP-UK	Tablet			null
10	14-OCT-UK	Hiking B			null
11	17-JUN-IN	Laptop			null
12	25-NOV-AU	Sneaker			null
13	08-DEC-DE	Coffee			null
14	18-FEB-CA	Smartw			null
15	16-APR-ES	headph			null
16	21-AUG-CA	Laptop			null
17	20-AUG-CN	Smartw			null
18	27-JAN-IT	Laptop	Asus	980	10
19	01-MAR-UK	Sunglasses	Oakley	250	15
20	14-AUG-US	Camping Tent	Coleman	null	10
21	14-MAY-RU	Camera	Nikon	700	50
22	09-JAN-CA	Microwave	Panasonic	80	20

Replace Values

Replace one value with another in the selected columns.

Value To Find: laptop

Replace With: Laptop

Advanced options

☐ Match entire cell contents

☐ Replace using special characters

Insert special character

OK Cancel

Query Settings

PROPERTIES

Name: Table1

APPLIED STEPS

Source

Changed Type

Removed Duplicates

Corrected inconsistent category

Replaced Value

Table1

Table.ReplaceValue(#"Corrected the Product name", "smartphone", "Smartphone", Replacer.ReplaceValue, {"Product Name"})

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
1	28-JAN-US	Laptop	Dell	1000	30
2	15-FEB-US	Sneaker			Electronics
3	03-MAR-US	Coffee			null
4	11-APR-US	Smartp			null
5	22-MAY-US	Backpak			null
6	07-JUN-UK	Headph			null
7	19-JUL-UK	T-shirt			null
8	23-AUG-UK	Blender			null
9	05-SEP-UK	Tablet			null
10	14-OCT-UK	Hiking B			null
11	17-JUN-IN	Laptop			null
12	25-NOV-AU	Sneaker			null
13	08-DEC-DE	Coffee			null
14	18-FEB-CA	Smartw			null
15	16-APR-ES	Headph			null
16	21-AUG-CA	Laptop			null
17	20-AUG-CN	Smartw			null
18	27-JAN-IT	Laptop			null

Replace Values

Replace one value with another in the selected columns.

Value To Find: smartphone

Replace With: Smartphone

Advanced options

☒ Match entire cell contents

☐ Replace using special characters

Insert special character

OK Cancel

Query Settings

PROPERTIES

Name: Table1

APPLIED STEPS

Source

Changed Type

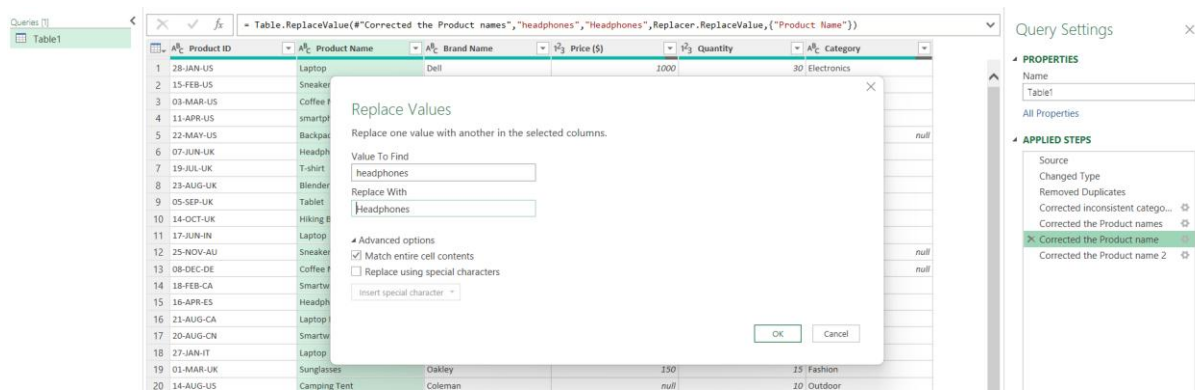
Removed Duplicates

Corrected inconsistent category

Corrected the Product names

Corrected the Product name

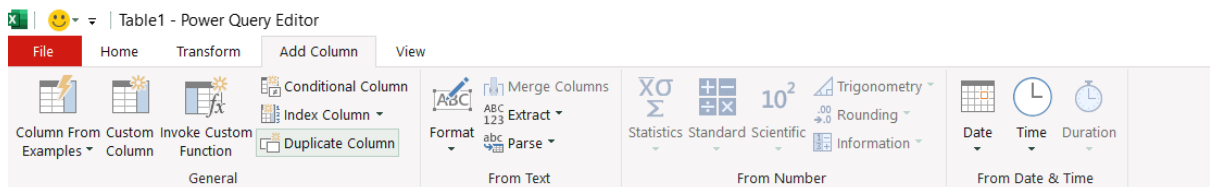
Corrected the Product name 2



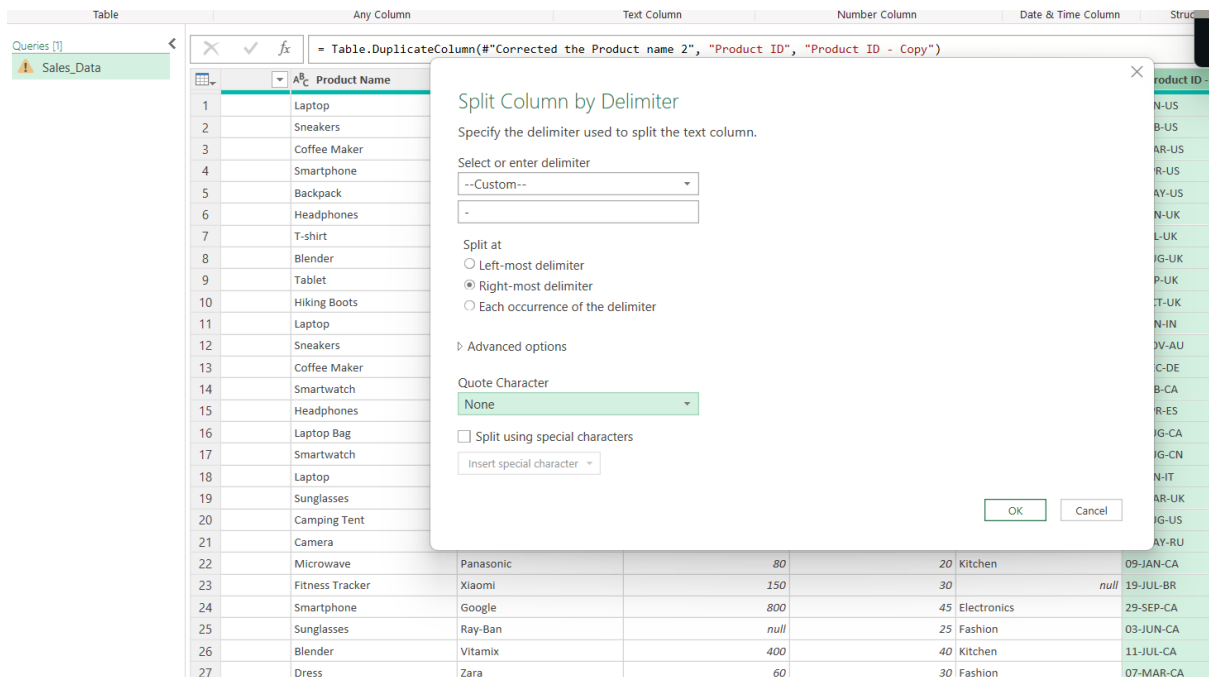
3. Splitting and Merging Data:

There is need of manufacturing date , country code. Therefore, separated this information from the Product ID by using Transform-> Split columns

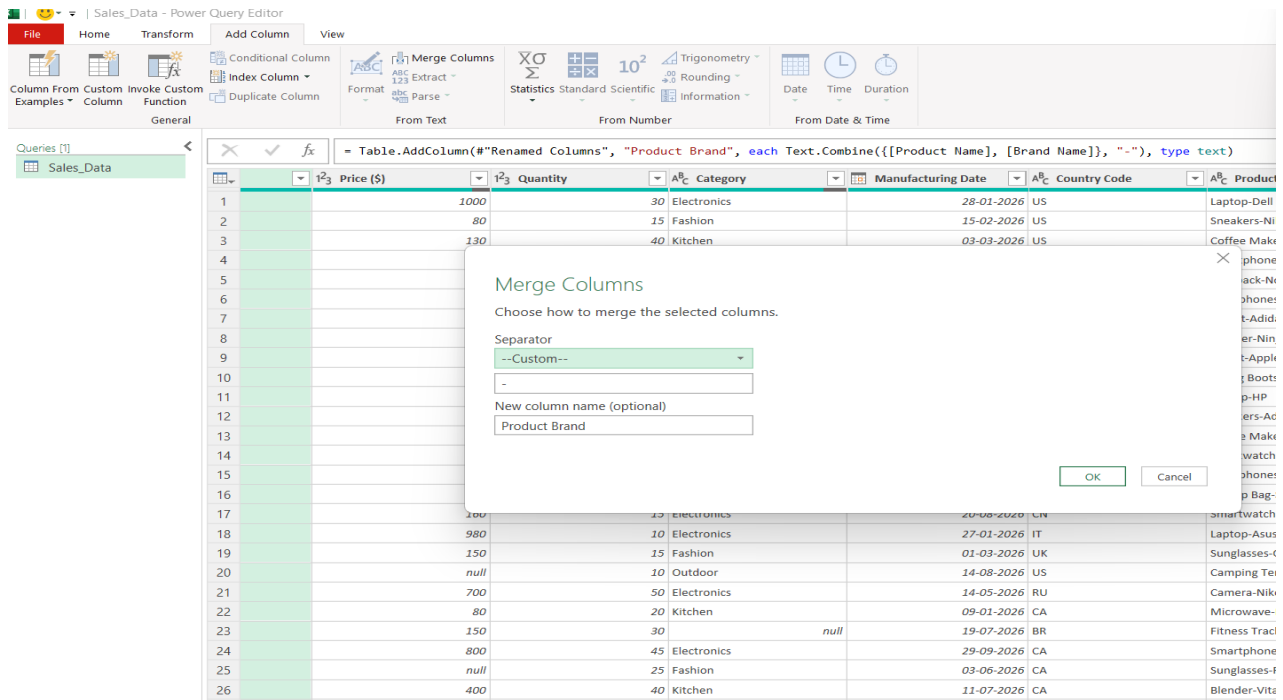
- Duplicated the column to convert by Add column option



- Selected the Split column by Delimiter, since the code and date are segregated by “-” delimiter



- Merged the Product name and Brand name, inserted new column as “Product Brand”



Power Query Editor - Sales_Data

Formula Bar: `= Table.AddColumn(#"Renamed Columns", "Product Brand", each Text.Combine([Product Name], [Brand Name]), "-"), type text)`

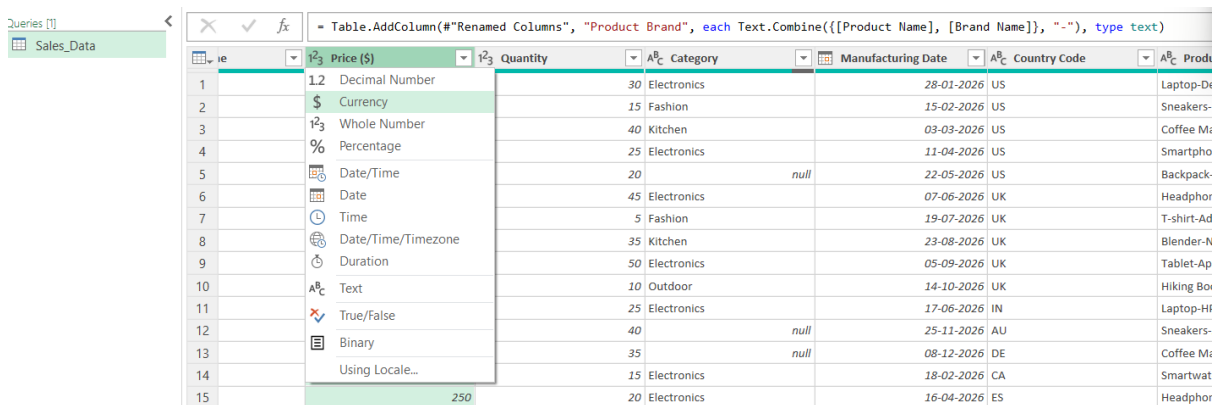
Table Columns: Price (\$), Quantity, Category, Manufacturing Date, Country Code, Product

Table Data:

	Price (\$)	Quantity	Category	Manufacturing Date	Country Code	Product
1	1000	30	Electronics	28-01-2026	US	Laptop-Dell
2	80	15	Fashion	15-02-2026	US	Sneakers-Ni
3	130	40	Kitchen	03-03-2026	US	Coffee Maki
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17	200	15	Electronics	20-08-2026	CN	smartwatch
18	980	10	Electronics	27-01-2026	IT	Laptop-Asus
19	150	15	Fashion	01-03-2026	UK	Sunglasses-t
20	null	10	Outdoor	14-08-2026	US	Camping Tei
21	700	50	Electronics	14-05-2026	RU	Camera-Niki
22	80	20	Kitchen	09-01-2026	CA	Microwave-
23	150	30		19-07-2026	BR	Fitness Trac
24	800	45	Electronics	29-09-2026	CA	Smartphone
25	null	25	Fashion	03-06-2026	CA	Sunglasses-t
26	400	40	Kitchen	11-07-2026	CA	Blender-Vit

4. Number Formatting:

- Already the manufacturing Date is formatted as “DD-MM-YYYY”. So , no need of changing it
- For the Price column , changed the format type by using format type option



Power Query Editor - Sales_Data

Formula Bar: `= Table.AddColumn(#"Renamed Columns", "Product Brand", each Text.Combine([Product Name], [Brand Name]), "-"), type text)`

Table Columns: Price (\$), Quantity, Category, Manufacturing Date, Country Code, Product

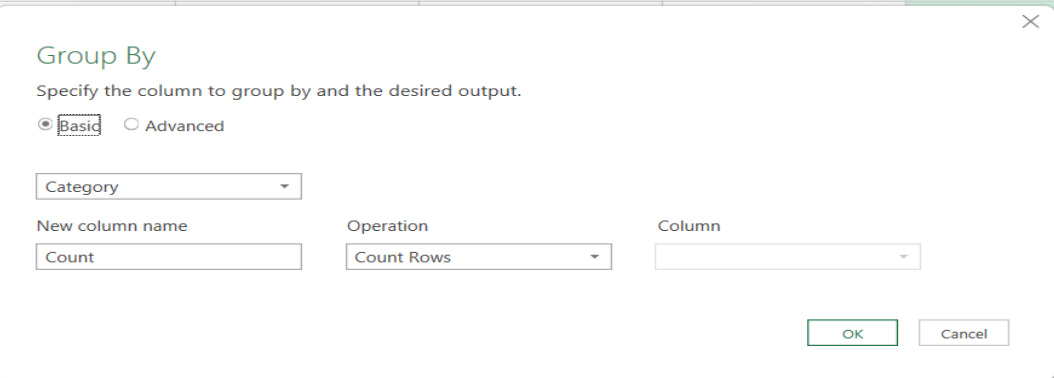
Table Data:

	Price (\$)	Quantity	Category	Manufacturing Date	Country Code	Product
1	\$ 1000	30	Electronics	28-01-2026	US	Laptop-Dell
2	\$ 80	15	Fashion	15-02-2026	US	Sneakers-Ni
3	\$ 130	40	Kitchen	03-03-2026	US	Coffee Maki
4		25	Electronics	11-04-2026	US	Smartpho
5		20		22-05-2026	US	Backpack-
6		45	Electronics	07-06-2026	UK	Headphor
7		5	Fashion	19-07-2026	UK	T-shirt-Ad
8		35	Kitchen	23-08-2026	UK	Blender-N
9		50	Electronics	05-09-2026	UK	Tablet-Ap
10		10	Outdoor	14-10-2026	UK	Hiking Bo
11		25	Electronics	17-06-2026	IN	Laptop-Hi
12		40		25-11-2026	AU	Sneakers-
13		35		08-12-2026	DE	Coffee Mi
14		15	Electronics	18-02-2026	CA	Smartwat
15		20	Electronics	16-04-2026	ES	Headphor

5. Handling Missing Values:

- Identified the null values in the Category and Price columns in power query

- Additionally, Excel functions can be used to determine the number of blank cells in each column and the overall count of blank values in the dataset
- Total of count the blank cells used formula=COUNTBLANK(Sales_Data[#All])
- To fill the missing values , we need to count by column wise ,
- Formula(s) used for the column wise blanks,
 - ✓ COUNTBLANK(Sales_Data[Product ID])
 - ✓ COUNTBLANK(Sales_Data[Product Name])
 - ✓ COUNTBLANK(Sales_Data[Brand Name])
 - ✓ COUNTBLANK(Sales_Data[Price (\$)])
 - ✓ =COUNTBLANK(Sales_Data[Quantity])
 - ✓ =COUNTBLANK(Sales_Data[Category])
- Filled the missing values by using the mode statistics option in the category column
 - ✓ Transform->Group by -> Category



	Category	Count
1	Electronics	12
2	Fashion	6
3	Kitchen	5
4	null	4
5	Accessories	2
6	Outdoor	2

- By mode Statistics, more often data points will be used for the null values. So filled the null values in category with Electronics
- **Alternative option:** If the data points are missed in most of columns means we can name as Unknown, else we can delete the row to reduce the error rate
- Close and loaded the data into Excel
- We can categorize the product name by similar products name
- By conditional formatting , we can find the duplicate values in the product name and change the category based on the existing similar Category

- Through that, Sorted the product name and matched the valid category for the products

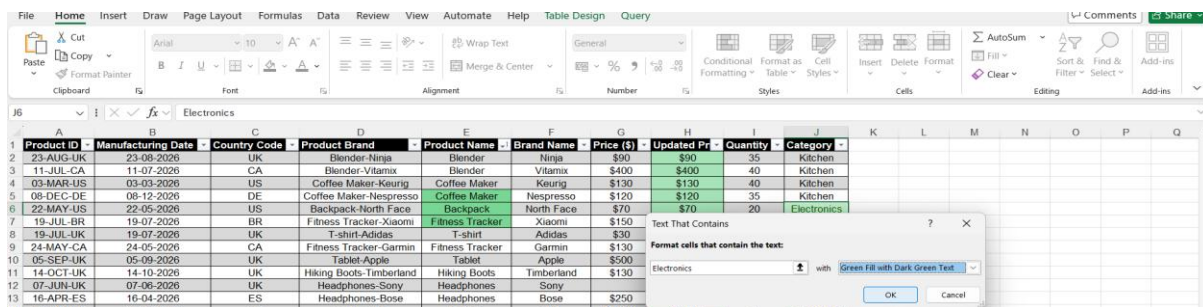
03-MAR-US	03-03-2026	US	Coffee Maker-Keurig	Coffee Maker	Keurig	130	40	Kitchen
08-DEC-DE	08-12-2026	DE	Coffee Maker-Nespresso	Coffee Maker	Nespresso	120	35	Electronics
15-FEB-US	15-02-2026	US	Sneakers-Nike	Sneakers	Nike	80	15	Fashion
25-NOV-AU	25-11-2026	AU	Sneakers-Adidas	Sneakers	Adidas	90	40	Electronics

- To Fill the Price value, the conditional functions are used. Since, the Product price have far outliers while comparing the category
- So , the average value for all the null values won't be suitable
- By using below formula, filled the null values in the price by category,

✓ IF(ISBLANK([@[Price(\$)]]),AVERAGEIF([Category],[@Category],\$G\$2:\$G\$32),[@[Price (\$)]])

6. Conditional Formatting:

- Highlighted the Electronics category by using the Conditional Formatting-> highlighted cell rules-> text that contains



- Using the conditional formatting, applied the data bar conditional formatting in the "Price" column
- Used the Conditional formatting , applied the icon set in the quantity column

